

## Do Cheaters Prosper? Effect of Externally Supplied Momentum During Resistance Training on Measures of Upper Body Muscle Hypertrophy

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### Abstract

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<https://doi.org/10.70252/GDBL2230> Exercise technique, defined as the controlled execution of bodily movements to ensure an exercise effectively targets specific muscle groups while minimizing the risk of injury, is a resistance training (RT) variable frequently highlighted as critical to successful RT program outcomes, with proper technique suggested to play a role in maximizing muscle development. This study examined the effects of externally applied momentum on RT-induced muscular adaptations in the upper extremities. Thirty young adults were recruited to participate in a within-participant design, with one limb randomly allocated to perform biceps curls and triceps pushdowns using strict form (STRICT) and the other using external momentum (CHEAT). Participants completed four sets of each exercise with 8-12 repetitions until momentary muscular failure, twice a week for eight weeks. We obtained pre-post proximal and distal measures of muscle thickness for the elbow flexors and extensors, and assessed circumference changes in the upper arms. Data were analyzed in a Bayesian framework including both univariate and multivariate mixed effect models with random effects. Differences between conditions were estimated as average treatment effects, with inferences based on posterior distributions and Bayes Factors (BFs). Results showed similar between-conditions increases for all muscle thickness sites as well as circumference measures, generating consistent support for the null hypothesis (BF = 0.06 to 0.61). Volume load was markedly greater for CHEAT compared to STRICT across each week of the intervention. In conclusion, the use of external momentum during single-joint RT of the upper extremities neither helped nor hindered hypertrophy of the target muscles.

**Keywords:** Exercise technique, cheat repetitions, exercise form, muscle growth

## Introduction

Resistance training (RT) has the potential to augment muscular hypertrophy through the manipulation of numerous variables, including training volume, proximity to muscular failure, and the execution of exercise techniques.<sup>1,2,3</sup> Exercise technique, defined as “the controlled execution of bodily movements to ensure an exercise effectively targets specific muscle groups while minimizing the risk of injury,”<sup>4</sup> is another variable frequently highlighted as critical to successful RT program outcomes, with proper technique suggested to play a role in maximizing muscle development.<sup>4</sup>

Key components of exercise technique, such as body positioning, range of motion and repetition tempo, have been individually studied.<sup>5,6,7</sup> Moreover, exercise technique guidelines are frequently based on biomechanical principles and applied anatomy, yet research directly examining the impact of these variables on hypertrophy is limited.<sup>4</sup> More specifically, an aspect of RT technique that lacks direct research is the use of external momentum during repetitions. External momentum, which often accompanies non-strict or “cheat” repetitions, involves ancillary muscle groups contributing to the movement. In contrast, strict repetitions aim to isolate the target muscle by minimizing momentum and additional muscle involvement.<sup>4</sup> For instance, performing a biceps curl with strict form limits hip or leg drive use, while cheat repetitions allow such contributions.

Cheat repetitions are frequently regarded as less effective than strict repetitions in promoting muscular adaptations.<sup>4</sup> To test this hypothesis, Arandjević<sup>8</sup> developed a biomechanical model that simulated the use of external momentum during the shoulder lateral raise exercise. Results from the computer simulation suggested that use of a “moderate” amount of momentum during the beginning of a repetition (initial angular velocities  $\sim 57.5^\circ \text{ s}^{-1}$ ) heightened abduction torque of the target muscles irrespective of increases in load. Moreover, results indicated that torque was further increased when external momentum was applied with a 25% increase in load. Based on a hypothesized interaction between the absolute magnitude of muscular tension and the duration of this tension in optimizing hypertrophy, the model suggested that incorporating moderate momentum at the beginning of each repetition provides the best balance between safety and stimulation of the target musculature. Despite these findings, excessive momentum could conceivably diminish the hypertrophic stimulus by reducing target muscle engagement or time spent at long muscle lengths, which could limit hypertrophic adaptations given the suggested importance of training at long muscle lengths for hypertrophy.<sup>9</sup>

To date, no studies have directly examined the effect of strict versus cheat repetitions on muscle hypertrophy. While theoretical and biomechanical perspectives provide some rationale for the use of strict or cheat repetitions, the practical implications of these approaches for optimizing muscular adaptations remain to be determined. Therefore, this study aimed to investigate the effect of externally applied momentum on RT-induced muscle hypertrophy of the upper body extremities in untrained men and women. We hypothesized that the use of external momentum would impair muscular adaptations due to a reduction in stimulus to the target musculature.<sup>4</sup>

## Methods

### Participants

Participants included 30 male and female volunteers recruited from a university population. Sample size determination was made based on considering the analysis approach (Bayesian analyses considering strength of evidence), previous research, the within participant design, and pragmatic considerations of ability to recruit between twenty to thirty participants. We employed a modified version of the workflow suggested by Wang and Gelfand<sup>10</sup> to quantify likely precision in our average treatment effect estimate across our potential sample sizes by investigating the width of the 95% credible intervals. We also performed simulation-based calibration of Bayes factors to assess whether the correct hypothesis was likely to be supported based on sample size and study design.<sup>11</sup> To assess likely precision, we first simulated prior predictive data for samples of size  $n$  using informative priors (sampling priors). These priors were based on results from meta-analyses investigating within group changes and comparative differences in strength and conditioning research.<sup>12,13</sup> Priors were set on a standardized scale and included a distribution describing typical improvement  $N(0.44, 0.40^2)$ , an average treatment effect  $N(0.30, 0.27^2)$ , a heterogeneous response  $N(0, 0.15^2)$  and measurement error  $N(0, 0.20^2)$ . The fitting priors used an average treatment effect prior of  $N(0, 0.40^2)$ . Simulation based calibration of Bayes factors assumed equal prior probabilities for the null ( $H_0$ ) and alternative ( $H_1$ ) hypotheses. Simulation and fitting was conducted with the neutral prior  $N(0, 0.40^2)$  and with the average treatment effect set to zero on half the iterations. Calibration was performed by examining the average posterior model probability (and determining if this matched the true 50%), and the average percentage of posterior allocated to the true hypothesis. Models were fit across  $m=500$  iterations for sample sizes  $n=20, 25$ , and  $30$ , with results presented in the supplementary material (Table S1).

To qualify for inclusion in the study, male and female participants were required to be: (a) between the ages of 18-40 years; (b) free from existing cardiorespiratory or musculoskeletal conditions that would impair the ability to lift weights and answer “no” to all items of the Physical Activity Readiness Questionnaire<sup>14</sup>; (c) untrained, defined as not having performed resistance-exercise for their upper body in the past year, and; (d) self-reported as free from consumption of anabolic steroids or any other illegal agents known to increase muscle size for the previous year. Participants were asked to refrain from consuming creatine products throughout the course of the study period since creatine has been shown to positively impact muscle growth when combined with RT.<sup>15</sup>

The participants were randomly assigned to 1 of 2 experimental unilateral, within-subject RT conditions for the elbow flexors and elbow extensors: traditional (STRICT:  $n=30$ ) or applied external momentum (CHEAT:  $n=30$ ). Randomization into groups was carried out using block randomization, with two limbs per block, via online software ([www.randomizer.org](http://www.randomizer.org)). Approval for the study was obtained from the college’s Institutional Review Board. Written informed consent was obtained from all participants before beginning the study. This research was carried out fully in accordance with the ethical standards of the *International Journal of*

*Exercise Science*.<sup>16</sup> The methods for this study were preregistered (<https://osf.io/5w3z9>) prior to recruitment.

### *Protocol*

The resistance exercise protocol included 2 exercises: unilateral supinated dumbbell biceps curl and unilateral pronated cable triceps pushdown. The protocol was performed on two non-consecutive days per week for 8 weeks. Each session consisted of 4 sets of 8-12 repetitions for each exercise. The resistance load was adjusted to ensure that participants stayed within the recommended repetition range. Research assistants supervised the participants during each session and provided verbal encouragement to perform sets until momentary muscular failure (the inability to perform another concentric repetition). Participants were instructed to abstain from performing any additional upper body RT outside of the study for the entire duration. To prevent non-localized fatigue from affecting limb performance, participants were given a 1-minute rest period between each arm when performing a given exercise and a 2-minute rest period when transitioning to the next exercise. Exercise order was not standardized, but participants completed all 4 sets of the first exercise before starting the next.

Before the training intervention, the participants underwent a 10-repetition maximum (RM) test for the biceps curl and triceps pushdown to determine the starting loads and familiarize them with the prescribed technique outlined in the study. To further aid visual learning, participants were provided with instructional videos to enhance their understanding (the videos can be viewed in the supplemental material). The RM testing was conducted according to the guidelines established by the National Strength and Conditioning Association (NSCA).<sup>17</sup>

The exercise instruction for the STRICT condition was consistent with guidelines provided by NSCA's Exercise Technique Manual for Resistance Training.<sup>18</sup> While this book provides instructions on specific aspects of exercise technique, it does not specify a prescriptive repetition tempo. We standardized the eccentric actions between conditions using a tempo of ~2 seconds as visually determined by the supervising researcher, which is consistent with general recommendations for the eccentric phase as well as for training to optimize muscle hypertrophy.<sup>19</sup> Given the research question, concentric tempo was specific to the condition, with STRICT performing controlled repetitions of ~1 second and CHEAT performing repetitions as explosively as possible.

### *STRICT RT instruction*

Participants adhered to the following performance guidelines for the STRICT condition. For the dumbbell biceps curl, participants began by positioning their unilateral arm alongside their torso, with palms supinated while standing with their feet shoulder – or hip width apart and their knees slightly flexed. Each repetition was executed through a full range of motion, consisting of complete extension and flexion at the elbow. Participants were instructed to maintain a stationary body position and to avoid swinging the torso (i.e., hyperextending the hips and spine), shrugging the shoulders, hyperextending the neck, extending the knees, or rising on the toes to assist in the concentric action.

For the cable triceps pushdown (Freemotion Fitness, Logan, UT, USA), participants maintained a pronated grip, upright torso, and 0 degrees of shoulder flexion/abduction ensuring elbow extension was the only movement occurring during the exercise.

#### *CHEAT RT instruction*

For the CHEAT condition, participants performed exercises with the use of external momentum during the concentric actions. As such, participants began the biceps curl movement by swinging the weight from full elbow extension at the bottom position to full elbow flexion at the top of the movement. To reinforce prescribed performance, they were verbally encouraged to utilize external momentum to lift the weight as often as possible until they reached failure.

For the triceps extensions, participants were instructed to use external momentum throughout the set until they could not fully extend the elbow. This included allowing the elbows to flare during each repetition, employing leg drive by bending the knees to assist in the downward motion, and assuming a forward lean to facilitate the completion of a full repetition.

#### *Measurements*

Muscle thickness (MT) was measured using ultrasound imaging as previously described.<sup>20</sup> In brief, a trained technician used a B-mode ultrasound imaging apparatus (Model ECO 3 Expert, CHISON Medical Technologies Co., Ltd., Wuxi, China) and applied a water-soluble transmission gel (Aquasonic 100 Ultrasound Transmission Gel, Parker Laboratories Inc., Fairfield, NJ) to a 4-12 MHz linear array ultrasound probe for all measurement sites. The probe was placed perpendicular to the tissue interface taking care not to depress the skin. Once a satisfactory image quality was achieved, the technician saved the image to a hard drive and measured MT by determining the distance from the subcutaneous adipose tissue-muscle interface to either the aponeurosis or the muscle-bone interface. Multiple measurements were taken at different points on the anterior and posterior upper arm because various sites have been suggested to be included when assessing muscle change.<sup>21</sup> Measurements of the elbow flexors and elbow extensors were taken at 55% (proximal) and 65% (distal) of the distance from the acromion process of the scapula to the lateral epicondyle of the humerus.

To ensure that muscle swelling from training did not obscure muscle thickness results, an image was obtained at least 48 hours after the participants' training sessions for both the pre-and post-study assessments. This protocol is consistent with research indicating that an acute increase in muscle thickness returns to baseline within 48 hours following a RT session.<sup>22</sup> We determined the intraclass correlation coefficients (ICC) and coefficients of variation (CV) in a cohort of 8 subjects for the assessed sites on 2 separate occasions separated by 24-48 hours; values for ICC and CV in these sites ranged from 0.97-0.99 and 1.19%-4.19%, respectively.

#### *Anthropometry*

To reduce the potential for confounding from lifestyle factors, participants were told to refrain from eating 12 hours before testing, eliminate alcohol consumption for 24 hours, and void their

bladder immediately before anthropometric testing. Each participant's height was measured using a stadiometer, and assessments of initial body mass and percent body fat was obtained by multifrequency bioelectrical impedance analysis (Model 770, InBody Corporation, Seoul, South Korea) as per the instructions of the manufacturer.

A three-dimensional optical scanner was then used to assess muscle circumferences of the upper arms. Participants stood on a rotating scale that performed a 360-degree scan of their body. This scan estimated circumferences, with a focus on the combined girth of the elbow flexors and extensors.

#### *Volume Load*

Volume load, calculated as (sets x repetitions x load [kg]), was manually recorded during each session. The values were transcribed to a spreadsheet and reported for each condition in both the biceps curl and triceps pushdown exercises.

#### *Dietary Adherence*

Participants were advised to maintain their customary nutritional regimen to avoid potential dietary confounding of results. Dietary adherence was assessed by self-reported 5-day food records (including at least 1 weekend day) using the MacroFactor mobile application (<https://macrofactorapp.com/>). Nutritional data were collected twice during the study: 1 week before the first training session (i.e., baseline) and during the final week of the training protocol. Participants were instructed to record all food items and the respective portion sizes consumed for the designated period of interest. Each food item was individually entered into the program, and the program provided relevant information as to total energy consumption, as well as the amount of energy derived from proteins, fats, and carbohydrates for each time period analyzed.

#### *Blinding*

To minimize potential bias, the researcher obtaining the MT measurements was blinded to group allocation, and the statistician performed blinded analyses for all outcomes. Moreover, group allocation was concealed from the researcher who determined whether a participant was eligible for inclusion.

#### *Statistical Analysis*

All analyses were conducted in R (version 4.4.0) within a Bayesian framework. Bayesian statistics represents an approach to data analysis and parameter estimation based on Bayes' theorem and can provide several advantages over frequentist approaches including formal inclusion of information regarding likely differences between intervention conditions based on knowledge from previous studies (e.g. through informative priors); and presentation of inferences based on intuitive probabilities.<sup>23</sup> Inferences were not drawn on within-condition change as this was not the focus of our research question,<sup>24</sup> although we descriptively presented within-group changes to help contextualize our findings. The effect of condition (STRICT vs

CHEAT) on outcome variables were estimated using linear mixed models with random effect structures included to account for the within participant design.<sup>25</sup>

All inferences were made from posterior distributions of model parameters describing estimates of the effect of intervention allocation and strength of evidence for the existence of a difference through Bayes factors. Informative prior distributions were used based on meta-analysis data on the specific research question and general strength and conditioning literature.<sup>13</sup> All analyses were performed using the R wrapper package brms interfaced with Stan to perform sampling.<sup>26</sup> A complete Bayesian workflow was adopted<sup>11</sup> that included prior predictive checks, posterior predictive checks, and simulation-based calibration of Bayes factors. To improve accuracy, transparency and replication of the analyses, the WAMBS-checklist (When to worry and how to Avoid Misuse of Bayesian Statistics) was used and reported.<sup>27</sup>

## Results

Twenty-five of the initial 30 volunteers completed the protocol (see Table 1). Reasons for drop-out are presented in Supplementary Figure S1. The average session attendance for participants who completed the protocol was 92%.

Table 1 presents the descriptive characteristics of participants. Table 2 presents a summary of pre- and post-intervention values for all outcomes. Table 3 presents estimates of average treatment effects (ATEs) between groups. Table 4 presents the raw weekly volume load data. Figure 1 presents standardized mean differences illustrating within-intervention changes.

**Table 1.** Descriptive characteristics of the participants.

| Variable               | Men (n = 11) | Women (n = 14) |
|------------------------|--------------|----------------|
| Height (cm)            | 172.2 ± 6.4  | 160.3 ± 6.4    |
| Body mass (kg)         | 75.8 ± 17.5  | 61.9 ± 15.7    |
| Age (yr)               | 20.2 ± 4.1   | 21.3 ± 3.6     |
| Bodyfat <sup>0</sup> % | 23.5 ± 11.8  | 32.6 ± 10.2    |

Several participants complained of a fast heart rate during performance of the CHEAT condition, presumably attributed to the cardiorespiratory component associated with swinging the weights, but these complaints subsided as they became accustomed to the technique. The majority of participants complained of muscle soreness during the initial weeks of the intervention but, consistent with the repeated bout effect<sup>28</sup>, this symptom did not persist after acclimation to the RT program. No other adverse events were reported during the intervention.

**Table 2.** Descriptive summary of pre- and post-intervention values for all outcomes

| Variable                       | STRICT (n=25)  |                |            | CHEAT (n= 25)  |                |            |
|--------------------------------|----------------|----------------|------------|----------------|----------------|------------|
|                                | Pre            | Post           | $\Delta\%$ | Pre            | Post           | $\Delta\%$ |
| <b>Biceps brachii 55 (mm)</b>  | 26.0 $\pm$ 5.0 | 27.5 $\pm$ 5.1 | 5.8%       | 26.1 $\pm$ 4.9 | 28.2 $\pm$ 5.3 | 8.0%       |
| <b>Biceps brachii 65 (mm)</b>  | 30.2 $\pm$ 4.9 | 31.9 $\pm$ 5.2 | 5.6%       | 30.3 $\pm$ 5.7 | 32.1 $\pm$ 5.7 | 5.9%       |
| <b>Triceps brachii 55 (mm)</b> | 29.3 $\pm$ 6.1 | 32.2 $\pm$ 6.3 | 9.9%       | 28.4 $\pm$ 6.4 | 31.4 $\pm$ 6.7 | 10.5%      |
| <b>Triceps brachii 65 (mm)</b> | 21.7 $\pm$ 5.2 | 24.9 $\pm$ 5.6 | 14.7%      | 21.5 $\pm$ 5.1 | 24.5 $\pm$ 4.9 | 14.0%      |
| <b>Circumference (cm)</b>      | 31.6 $\pm$ 5.3 | 32.1 $\pm$ 4.9 | 1.6%       | 32.0 $\pm$ 4.9 | 32.6 $\pm$ 4.9 | 1.9%       |

### *Muscle Thickness*

Results for both the elbow flexors and extensors were consistent with evidence supporting the null hypothesis. For the elbow flexors the support was ‘anecdotal’ at 55% (Bayes factor = 0.61) and ‘moderate’ at 65% (Bayes factor = 0.13) of the distance between the lateral epicondyle of the humerus and the acromion process of the scapula (Table 3). Multivariate analysis of the two regions of the elbow flexors provided ‘strong’ evidence supporting the null hypothesis (Bayes factor = 0.07). Univariate analyses for the elbow extensors provided ‘moderate’ support (Bayes factor = 0.24 and 0.25) for the null hypothesis at 55% and 65% of the distance between the lateral epicondyle of the humerus and the acromion process of the scapula, respectively (Table 3). Multivariate analysis of the two regions of the elbow extensors provided ‘strong’ evidence supporting the null hypothesis (Bayes factor = 0.06).

Within-condition analyses using standardized mean difference estimates indicated that interventions were likely to produce small to medium increases in muscle thickness for the elbow flexors and medium to large increases for the elbow extensors (see Figure 1). Output from the WAMBS checklist presented in the supplementary file identified no concerns with the analyses.

### *Arm Circumference*

Results for measurement of arm circumference provided ‘moderate’ support (Bayes factor = 0.18) for the null hypothesis (Table 3). Within-condition analyses using standardized mean difference estimates indicated that interventions were likely to produce small increases in arm circumference (see Figure 1). Output from the WAMBS checklist presented in the supplementary file identified no concerns with the analyses.

**Table 3.** Univariate and multivariate analyses of potential group pre to post differences for hypertrophy outcomes.

| Variable                 | Univariate Difference STRICT:CHEAT (95%CrI) | Bayes factor for univariate analysis with interpretation | Bayes factor for multivariate analysis with interpretation |
|--------------------------|---------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|
| Biceps brachii 55% (mm)  | 0.49 (-0.08 to 1.1)                         | 0.61 (Anecdotal support of H <sub>0</sub> )              | 0.07 (Strong support of H <sub>0</sub> )                   |
| Biceps brachii 65% (mm)  | 0.08 (-0.46 to 0.60)                        | 0.13 (Moderate support of H <sub>0</sub> )               |                                                            |
| Triceps brachii 55% (mm) | 0.06 (-1.1 to 1.2)                          | 0.24 (Moderate support of H <sub>0</sub> )               | 0.06 (Strong support of H <sub>0</sub> )                   |
| Triceps brachii 65% (mm) | -0.08 (-1.1 to 0.96)                        | 0.25 (Moderate support of H <sub>0</sub> )               |                                                            |
| Circumference (cm)       | 0.11 (-0.60 to 0.79)                        | 0.18 (Moderate support of H <sub>0</sub> )               | -                                                          |

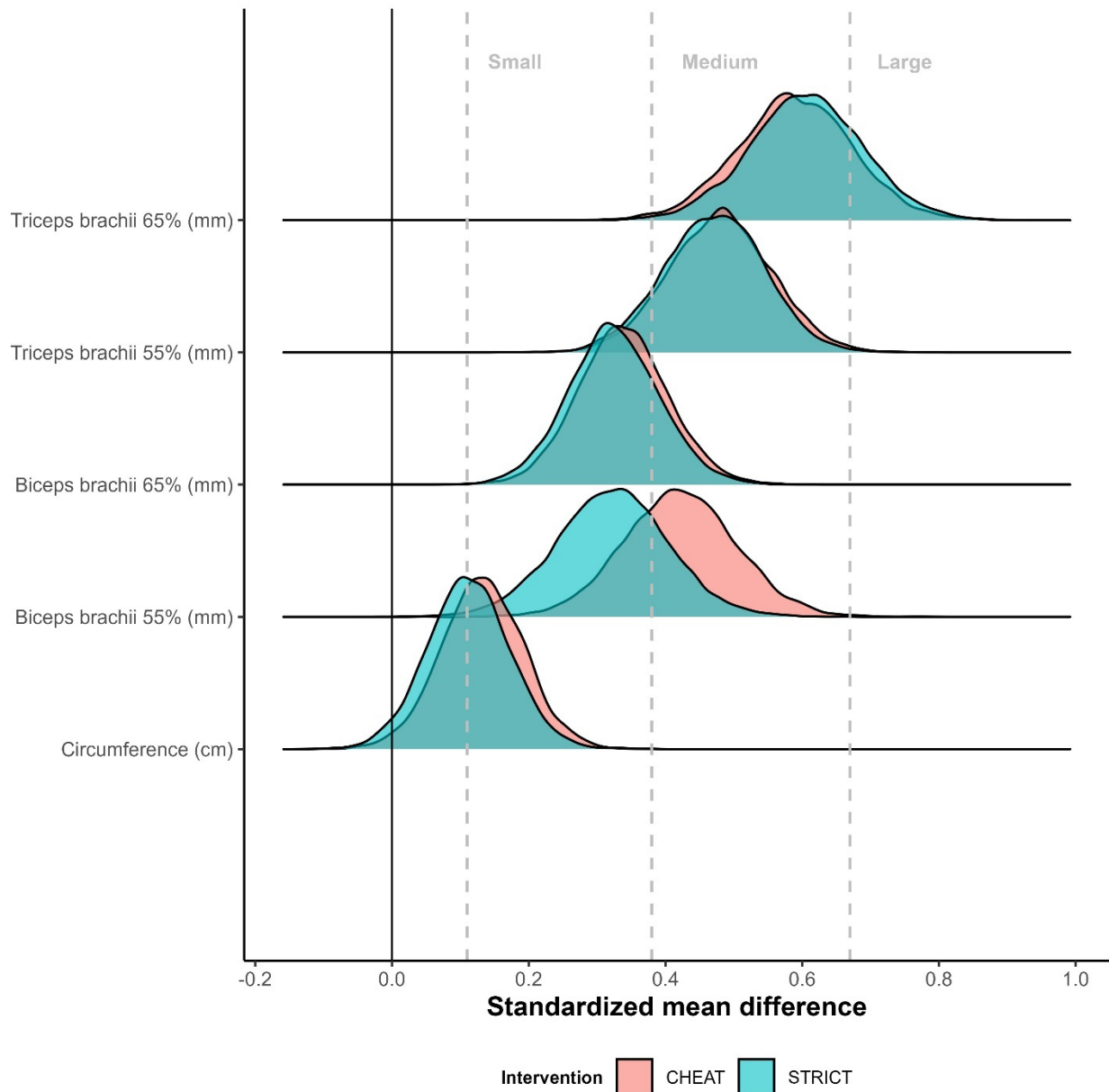
Positive values of group differences favor the CHEAT condition. CrI: Credible interval

### Volume Load

Total volume load at the beginning of the intervention was nearly double in the CHEAT condition compared to STRICT (Table 4). Total volume load increased for both conditions over the 8-week program, with CHEAT demonstrating greater increases in weekly volume (70.9% [95%CrI: 53.6 to 88.4]) compared to STRICT (35.7% [95%CrI: 18.3 to 53.0]; Supplemental Figure S2).

**Table 4:** Weekly volume load for each condition (kg).

|        | BiCheat | TriCheat | BiStrict | TriStrict |
|--------|---------|----------|----------|-----------|
| Week 1 | 22751   | 16879    | 11641    | 9097      |
| Week 2 | 22339   | 18630    | 10850    | 9702      |
| Week 3 | 21866   | 19260    | 11159    | 10586     |
| Week 4 | 20354   | 17166    | 11613    | 8469      |
| Week 5 | 24680   | 21834    | 12822    | 10030     |
| Week 6 | 24852   | 22215    | 12675    | 11200     |
| Week 7 | 24558   | 21552    | 13131    | 10786     |
| Week 8 | 18228   | 16365    | 9872     | 8070      |



**Figure 1.** Standardized mean differences illustrating within-intervention changes.

## Discussion

To our knowledge, this is the first study to directly investigate the effects of external momentum during RT on muscle hypertrophy. Our findings indicate that incorporating external momentum had no discernible effect on MT or upper arm circumference when performing an 8-week program of biceps curls and triceps pushdowns compared to the same exercises performed with strict technique. Results were consistent between conditions across all

outcomes, providing support for the null hypothesis. However, within-condition analyses showed that both interventions led to improvements over the training period. We hypothesized that the use of external momentum would reduce the stimulation of target muscles, thereby attenuating hypertrophic adaptations. However, this hypothesis was not supported, as hypertrophy outcomes were comparable between conditions.

Our findings are at odds with those of Arandjevićlo,<sup>8</sup> whose biomechanical model suggested the use of momentum during shoulder abduction heightened torque of the deltoid, conceivably promoting a greater hypertrophic stimulus in this muscle. Discrepancies between the two studies can be attributed to the limitations of extrapolating results of a biomechanical model into practice. By nature, a model is only as good as its underlying assumptions. It is therefore possible that some of the assumptions in Arandjevićlo's model<sup>8</sup> may not accurately approximate the complexity and variability of in vivo biological systems,<sup>29</sup> which could have influenced the ability to draw conclusions. Moreover, Arandjevićlo's model<sup>8</sup> was specific to humeral abduction for targeting the deltoid; it is possible that this model may not translate to the use of external momentum in other joint actions and skeletal muscles. Alternatively, even though our study had sufficient statistical power to detect probabilistic differences between conditions with reasonable confidence, we cannot rule out the possibility of a type II error. Thus, replication of our methods as well as investigation of the use of external momentum in other muscles is required to draw stronger inferences on the topic.

As expected, volume load was substantially greater when employing external momentum, with weekly values generally more than double that of STRICT. Volume load has been proposed to be a contributing factor in hypertrophy,<sup>30</sup> although this theory remains controversial.<sup>31</sup> On the surface, our results would seem to refute the effects of volume load on muscle development. However, given that the use of external momentum involves body segments other than the agonists, it is likely that some, if not much, of the muscular stress during training was redirected away from the target musculature. Thus, no strong conclusions can be drawn in this regard.

Of note, volume load continued to increase over the course of the first 7 weeks of the study for both conditions but decreased precipitously in the final week. The volume load drop-off can largely be attributed to the need for several of the participants to post-test before they completed their final training session due to scheduling conflicts and the impending holiday break.

From a practical standpoint, our findings can be interpreted in several ways. On one hand, the lack of difference in measures of hypertrophy between conditions conflicts with the claim that strict technique is crucial to optimizing muscle development.<sup>4</sup> This could be taken to mean that trainees can simply focus on lifting weights with little regard to generally accepted kinematic principles, at least when performing single-joint upper limb exercises. Conceivably, the ability to use heavier loads when performing repetitions with external momentum may offset the dispersion of forces to other body segments, ultimately inducing similar levels of mechanical tension in the target muscles as when training with strict technique. If true, this would help to explain the similar adaptations between conditions. We only measured hypertrophy of the upper arm muscles, but it is possible that other muscles involved in the CHEAT condition also may have hypertrophied from the ancillary stress. Future studies should investigate the effect

of external momentum on adaptations of non-target muscles to determine if this strategy may represent a time-efficient approach for overall muscle development.

On the other hand, using external momentum throughout a training program may increase injury risk to soft tissue structures including muscles, tendons and ligaments.<sup>8</sup> Direct evidence investigating the effects of technique on injury rate remains scant due to the inherent issues associated with objectively investigating the topic. While no injuries were reported during the course of our study, the 8-week interventional period and limited number of participants may have been insufficient to adequately assess the topic. Although speculative, continually subjecting soft tissues to excessive external momentum could be deleterious to these structures. Factors such as the amount of external momentum used during exercise performance and the specific body segments involved (e.g., spine, hips, etc.) may play a role in the associated risk of the strategy. Thus, trainees should take into account the potential increased injury risk from a cost-benefit standpoint when deciding how to perform a given exercise.

While the present study provides novel insights into the effects of RT techniques on measures of muscle development, there are several limitations that must be considered when drawing practical inferences on the topic. First and foremost, this was a proof-of-principle study designed to isolate the effects of external momentum on muscular development. To exert maximal control over the training stimulus to the biceps and triceps, we limited exercise selection to elbow flexion and elbow extension as the performance of other upper body exercises (e.g., chest presses, rows, etc.) would have directly influenced hypertrophy of the target muscles. This necessitated the use of untrained individuals since those engaged in regular RT likely would not have consented to only perform exercises for the arms over the 8-week intervention. Therefore, our findings cannot necessarily be extrapolated to those with RT experience; future research should investigate the topic in this population under more ecologically valid protocols. Second, the CHEAT condition used external momentum throughout all repetitions. Conceivably, there may be a benefit to employing external momentum after performing strict repetitions at or close to failure as often performed by bodybuilders. This would essentially be a form of training “beyond” failure, which has been shown to be effective via the extended use of partial repetitions at longer muscle lengths after training to fatigue with a full range of motion.<sup>32</sup> This hypothesis warrants future investigation. Third, it was difficult to standardize form in CHEAT; therefore, we cannot rule out the possibility that between-participant differences in the amount of momentum and the kinematics associated with performance in this condition may have influenced results. Finally, our results are specific to hypertrophy of the elbow flexors and extensors when performing arm curls and triceps pushdowns; the findings cannot necessarily be extrapolated to multi-joint movements and other muscle groups.

The use of external momentum during single-joint RT of the upper extremities neither helped nor hindered hypertrophy of the target muscles in untrained individuals. These results were observed even though the CHEAT condition performed approximately double the volume load of the STRICT condition. The potential increased injury risk associated with persistent use of excessive external momentum requires consideration when deciding on technique for a given exercise. Future research should investigate the use of external momentum to extend sets after

fatiguing muscles with strict form, as well as assessing its effects in multi-joint exercises and with other muscle groups, especially in resistance trained populations.

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Competing Interests: BJS formerly served on the scientific advisory board of Tonal Corporation, a manufacturer of fitness equipment. The other authors have no competing interests to report in concert with this manuscript.

Data and Supplementary Material Accessibility: Data can be requested from the corresponding author upon reasonable request. Supplementary material are available on the Open Science Framework project page: <https://osf.io/jxaqn/files/osfstorage>

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